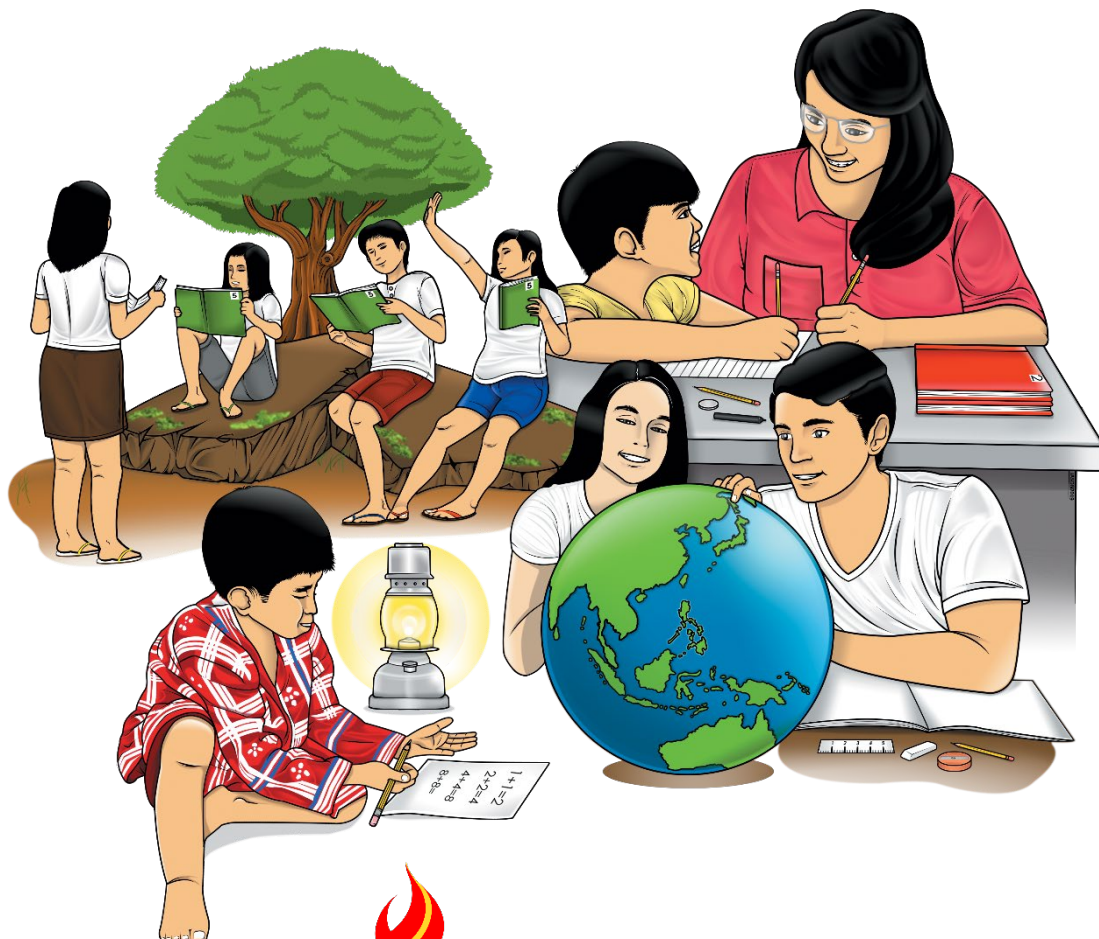


Science

Quarter 2 – Module 4

Animals: Characteristics of Vertebrates



Science – Grade 6
Alternative Delivery Mode
Quarter 2 – Module 4: Animals: Characteristics of Vertebrates
First Edition, 2020

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Science

Quarter 2 – Module 4: Animals: Characteristics of Vertebrate

Introductory Message

This Self-Learning Module (SLM) is prepared so that you, our dear learners, can continue your studies and learn while at home. Activities, questions, directions, exercises, and discussions are carefully stated for you to understand each lesson.

Each SLM is composed of different parts. Each part shall guide you step-by-step as you discover and understand the lesson prepared for you.

Pre-tests are provided to measure your prior knowledge on lessons in each SLM. This will tell you if you need to proceed on completing this module or if you need to ask your facilitator or your teacher's assistance for better understanding of the lesson. At the end of each module, you need to answer the post-test to self-check your learning. Answer keys are provided for each activity and test. We trust that you will be honest in using these.

In addition to the material in the main text, Notes to the Teacher are also provided to our facilitators and parents for strategies and reminders on how they can best help you on your home-based learning.

Please use this module with care. Do not put unnecessary marks on any part of this SLM. Use a separate sheet of paper in answering the exercises and tests. And read the instructions carefully before performing each task.

If you have any questions in using this SLM or any difficulty in answering the tasks in this module, do not hesitate to consult your teacher or facilitator.

Thank you.



What I Need to Know

This module was designed and written with you in mind. It is here to help you master the lesson about animals. The scope of this module is used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

This module is about:

- Describing the distinguishing characteristics of each group of vertebrates
- Classifying each group of vertebrates
- Listing examples of each group of vertebrates

After going through this module, you are expected to:

1. Describe the distinguishing characteristics of each group of vertebrates
2. Classify each group of vertebrates
3. List examples of each group of vertebrates



What I Know

Directions: Match the characteristics of vertebrates in column **A** with their classifications in column **B**. Write the letters of the correct answer in your Science journal.

A

1. They have mammary gland to feed their young.
2. They have moist skin and can live on land and in
3. They have dry scaly skin.
4. They have beak and wings used in flying.
5. They have gills and fins and they live in water.

B

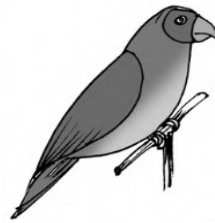
- A. birds
- B. amphibians
- C. mammals
- D. fish
- E. reptiles

Directions: Identify the classification of animals by writing mammals, amphibians, fish, birds, or reptiles on the blanks provided. Write your answers in your Science journal.

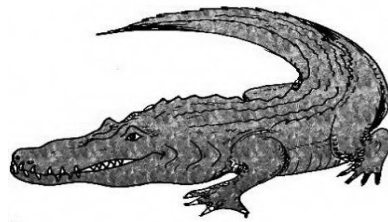
6. _____



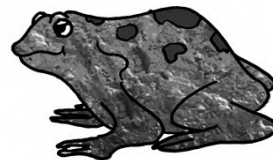
7. _____



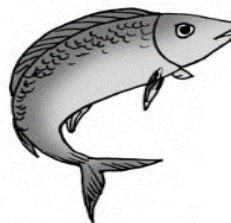
8. _____



9. _____



10. _____



Lesson 1

Animals: Characteristics of Vertebrates

Animals can be found anywhere in our environment: in the sea, grassland, forests, ponds, sanctuary, and even in our homes. These animals in different ecosystem and habitat are classified as vertebrates and invertebrates. Vertebrates are animals with backbones and have varying body parts used in walking, swimming, flying, climbing, and hopping. They eat grass, meat, insects, and seeds as their food.

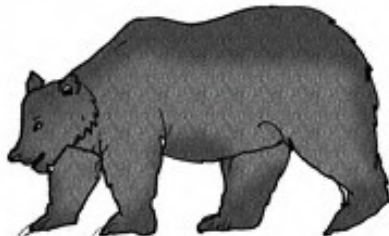


What's In

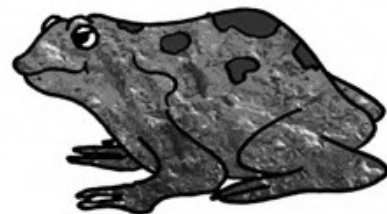
Direction: Choose the letters of the correct answer and write your answers in your Science journal.

1. Which of the following animals can be found in the locality?

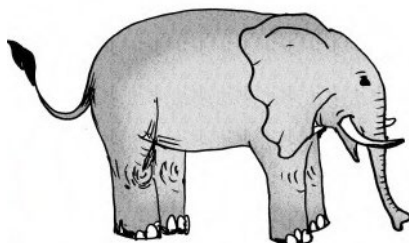
A.



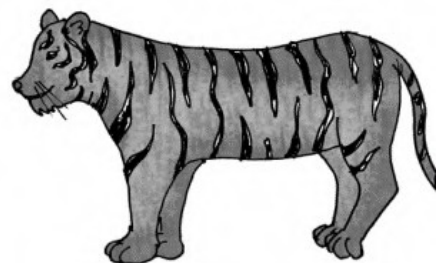
C.



B.



D.



2. Animals are classified as vertebrates and invertebrates. Vertebrates are animals with backbones while invertebrates are animals without backbones.

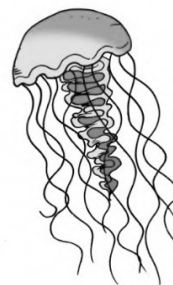
Directions: Identify the following animals whether vertebrates and invertebrates. Write your answers in your Science journal.

- A. butterfly - _____
- B. carabao - _____
- C. chicken - _____
- D. snail - _____
- E. praying mantis - _____

3. To what group of animals does the illustration belong?

A. vertebrates C. invertebrates

B. arthropods D. insects



4. Which is the distinguishing characteristic of invertebrates?

- A. They have backbones.
- B. They have spinal cord.
- C. They have no backbones.
- D. They have brain.



What's New

Read the poem and answer the questions that follow.

Vertebrates

Have you ever seen, animals that swim?
They have gills, fins, and scales
Spotted in the sea, oceans, and rivers
These are fish like tilapia, swordfish, and milkfish.

Have you ever seen animals that crawl?
They have dry-scaly skin just like snakes, lizards, and turtles
Some have four legs, others have none
These are reptiles hatched from eggs.

Have you ever seen four-legged animals that hop and jump?
They have fur, some have hairs, others have skin
They give birth, feed their young with their mammary glands
They are mammals like dogs, cats, and goats.

Have you ever seen animals that fly? They have wings, beak, and claws
Living in trees and forests, building their nests in trunks
They are birds like parrots, owls, and doves

Have you ever seen animals that croak?
They have smooth scaly skin living on land and in water
They love to hop and eat insects at night
These are amphibians just like salamanders, toads, and frogs.

By: Judy C. Villanueva

Directions: Complete the missing letters to form the correct answer. Write your answers in your Science journal.

1. What is common among animals that swim?
G _ _ _ s
2. What characteristic is common among reptiles?
C _ _ w _
3. What is common among parrots, owls, and doves?
W _ _ _ s
4. What is common among mammals like dogs, cats, and goats?
F _ _
5. What is common among amphibians?
S _ _ _ t h s _ _ l _ skin

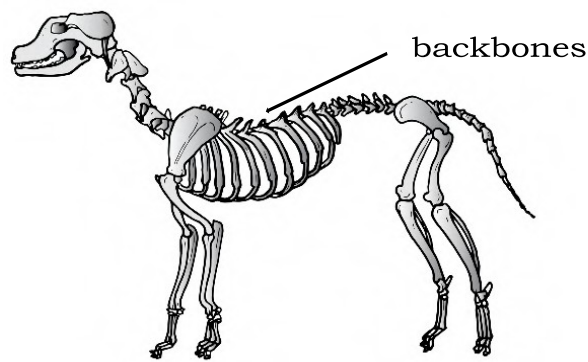


What is It

Animals found in the locality are classified into vertebrates and invertebrates. Vertebrates are animals with backbones that give shape and support their body. Bones are predominant feature in the body of vertebral animals that serve as protection of their internal organs. The presence of bones in the animal's body helps the animals to walk, jump, crawl, and swim.

Animals can be oviparous and viviparous. **Viviparous** are animals that are born alive by their parents such as dogs, carabao, and cats, while **oviparous** animals are those that are hatched from egg such as chicken, duck, and snake.

Figure 1: Animal with backbone



Vertebrates are group into the following:

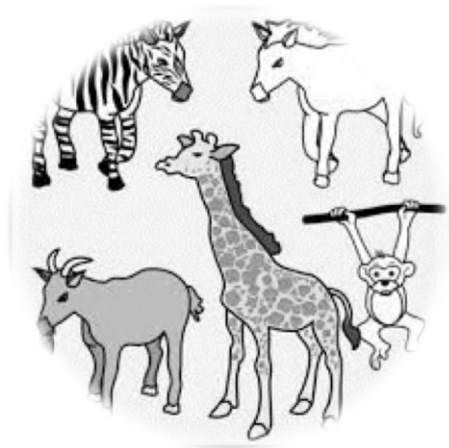


Figure 2: Mammals

Mammals

Mammals are warm- blooded animals that are born alive or viviparous. They have hair or fur as their body covering and mammary gland to feed their young with their milk.

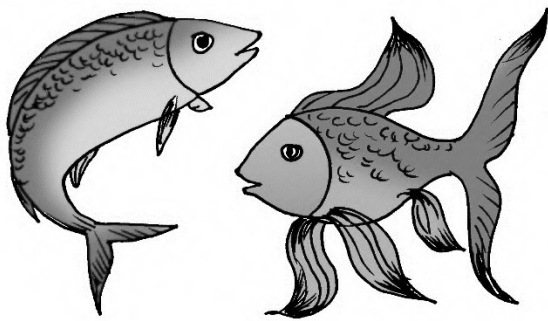


Figure 3: Fish

Fish

Fish are cold-blooded animals covered with scales. They have tails and fins that help them swim. They are animals hatched from egg and breathe through their gills.

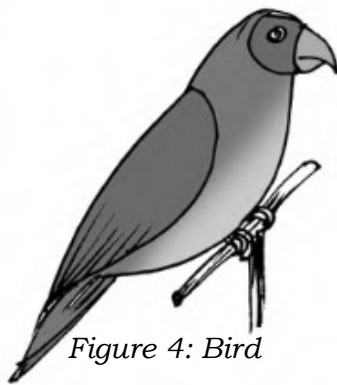


Figure 4: Bird

Birds

Birds are warm-blooded animals hatched from egg or oviparous. They have a pair of limbs for hopping and a pair of wings used in flying. There are birds who have wings but can't fly. A beak or bill is used in getting food as well as for protecting themselves from their enemies.

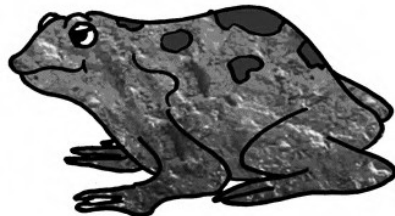


Figure 5: Amphibian

Amphibians

Amphibians are cold-blooded animals hatched from egg. They spend part of their lives in water and part on land. They have moist and scaleless skin. Some examples are salamander, newt, toad and caecilian.

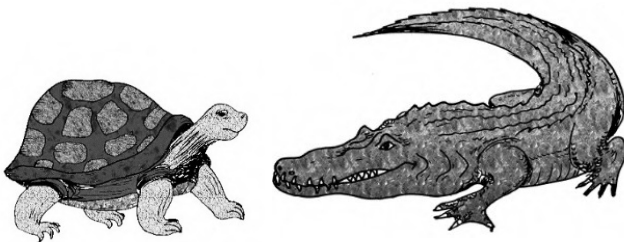
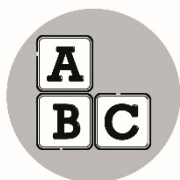


Figure 6: Reptiles

Reptiles

Reptiles are cold-blooded vertebrates hatched from egg. They have dry scaly skin and some have shells. Some examples are turtles, chameleon, snake, lizard, and gecko.



What's More

Directions: Classify the following vertebrates in their proper column. Write your answer in your Science journal.

bangus

maya

frog

lizard

cat

galonggong

snake

swamp frog

gecko “tuko”

bat

stingray

crocodile

tilapia

toad

carabao

Mammals	Amphibians	Reptiles	Birds	Fishes



What I Have Learned

Directions: Fill in each blank with the correct word. Write the answers in your Science journal.

I learned that...

Vertebrates are classified as, _____, _____, _____, _____, and _____.

The _____ are animals that are born alive and have hair or fur as their body covering. The _____ are animals that have scales, fins, and gills and live in water.

While _____ are animals that have dry scaly skin and hatched from egg. The _____ are animals that have moist skin and can live both on land and in water and _____ are animals with beak or bills and have wings.



What I Can Do

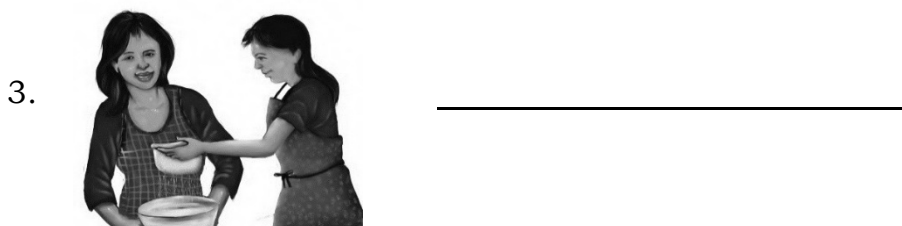
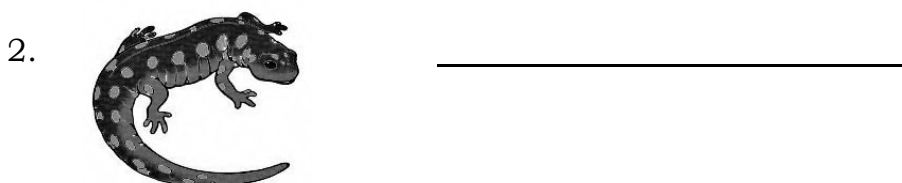
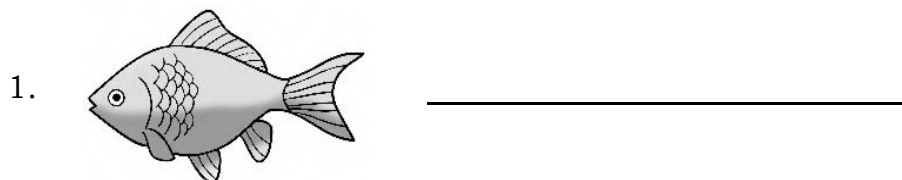
Directions: Answer the following questions. Write your answer in your Science Journal.

1. Frogs are useful animals that eat insects that are harmful to humans and other animals. People living in rural area exploit them by consuming them as food. Since frog population is already diminishing in number, in what way can you be of help to remind the people of the importance of frogs?
2. Mammals such as dogs and cats are cared as pet. They are furry animals that usually have parasitic ticks and fleas in their bodies. These parasites harm their host and often cause health problems to animals. How will you get rid of these pests that threaten your pet?

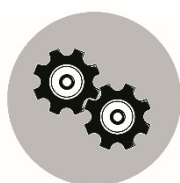


Assessment

Directions: Write the correct classification of vertebrates as illustrated. Write the answers in your journal.



6. All given set of animals are classified in one vertebrate group, which is NOT?
- gecko, snake, crocodile, iguana
 - dolphin, cat, horse, sheep
 - ostrich, bat, penguin, duck
 - toad, newt, salamander, frog
- A. I B. II C. III D. IV
7. Goats, cats, dogs, and rabbits are mammals have characteristics that differ from other vertebrates. Which of the following features does not describe a mammal?
- Mammals have mammary glands.
 - Mammals have scales and feathers.
 - Mammals have fur or hair.
 - Mammals are warm-blooded animals.
8. In what way do amphibians differ from reptiles?
- Amphibians have backbones.
 - Amphibians can live both on land and in water.
 - Amphibians can crawl.
 - Amphibians are oviparous.
9. Ostriches are classified as birds but cannot fly. Why?
- They have thick feathers.
 - They have heavy weight.
 - They are afraid of heights.
 - They prefer to be in land.
10. Some animals were taken from their natural habitat by hunters or people and were being sold in pet shops. Which of these will you do in order to minimize these activities that endanger different species of animals?
- Help the hunters to find a pet shop to sell their catch.
 - Report illegal activity to a Forest Ranger or authority.
 - Buy the animal being sold to be your pet.
 - Take care of the animal for a while and then sold it.



Additional Activities

Directions: Make an inventory of vertebrates found in your community and suggest ways of protecting these animals. Write your answer in your Science journal.

Example:

Name of Animals	Ways of Protecting these Animals
1.	
2.	
3.	



Answer Key

Assessment 1. Fish 2. Amphibian 3. Mammal 4. Bird 5. Reptile 6. C 7. B 8. B 9. B 10. B Additional Activities Answer may vary	What I Have Learned mammals, reptiles, amphibians, birds and fish (in any order) -Mammals -Fishes -Reptiles -Amphibians -Birds What I Have Learned Answer may vary	What's More Mammals: bat, cat, carabao Amphibians: swamp frog, toad, frog Reptiles: lizard, snake, crocodile, gecko Birds: Maya Fish: bangus, tilapia, stingray, galunggong
What's New 1. gills 2. crawl 3. wings 4. fur 5. smooth scaly	What's In 1. C 2. A. invertebrates B. vertebrates C. vertebrates D. invertebrates E. invertebrates 3. C 4. C	What I Know 1. C 2. B 3. E 4. A 5. D 6. mammals 7. birds 8. reptiles 9. amphibians 10. fishes

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